Problem A
A. Nihilation

You are given an array $A = [A_1, A_2, \ldots, A_N]$ consisting of N positive integers.

In one operation, you can choose integers m and k such that $1 \leq k < m \leq 10^9$ and set $A_i := (A_i \times k) \bmod m$ for $1 \leq i \leq N$.

What is the minimum number of operations needed to make all A_i equal to 0? Output any sequence of operations to be done. It can be proven that it is always possible to make all A_i equal to 0.

Input

Input begins with an integer N ($1 \leq N \leq 100\,000$). The next line contains N integers A_i ($1 \leq A_i \leq 10^9$) representing the given array A.

Output

In the first line, output the minimum number q of operations needed.

In the next q lines, output two integers m and k, representing the operation in the sequence of operations that makes all A_i equal to 0. If there are multiple such sequences, output any one of them.

Sample Input 1	Sample Output 1
5 4 1 2 6 3	2 12 6 3 2

Explanation of Sample 1: The following describes the sequence of operations done in the sample output.

1. $A_i := (A_i \times 6) \bmod 12 \implies A = [0, 6, 0, 0, 6]$
2. $A_i := (A_i \times 2) \bmod 3 \implies A = [0, 0, 0, 0, 0]$

It can be shown that no sequence of operations with length 1 can make all A_i equal to 0.

Sample Input 2	Sample Output 2
2 9 9	1 3 1

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Problem B
Burning Blocks

William and his friends are on a camping trip and have gathered some wooden blocks. Before nightfall, they stack these wooden blocks into N stacks from left to right. The i-th stack has W_i wooden blocks stacked on top of each other.

Each wooden block takes exactly one minute to burn. Once a wooden block is completely burnt, the fire will spread to all wooden blocks adjacent to it (the blocks immediately to its left, right, above, and below) and start burning them.

William starts burning from the outermost blocks, i.e. the blocks whose either left, right, or above side contains no blocks. Determine the total number of minutes required for all of the wood blocks to be completely burnt.

Input

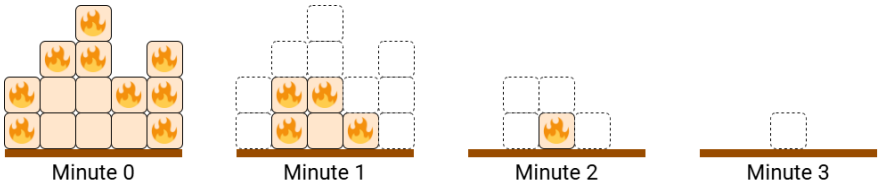
The first line contains an integer N ($1 \leq N \leq 200\ 000$). The second line contains N integers representing W_i ($0 \leq W_i \leq 10^9$), the number of wooden blocks in each stack.

Output

A single line representing the number of minutes required for all blocks to be completely burnt.

Sample Input 1	Sample Output 1
5 2 3 4 2 3	3

Explanation of Sample 1: The following illustrates the burning process.



Sample Input 2	Sample Output 2
13 2 5 6 6 4 3 4 3 1 2 4 8 4	4

Sample Input 3	Sample Output 3
3 1 0 2	1

Sample Input 4	Sample Output 4
1 0	0

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Problem C
Binary Grid

You are given an array A containing N non-negative integers. You have to construct a binary grid of size $N \times M$ such that:

- for any cell with value 1, any of its adjacent cells, i.e. cells that share one of its four sides, must **not** have the value 1.
- the number of cells with value 1 in row i is exactly A_i.

Find any such binary grid or tell that it's impossible to construct.

Input

The first line contains two integers N and M ($1 \leq N, M \leq 1000$). The next line contains N integers containing A_i ($0 \leq A_i \leq M$).

Output

Output N lines, each containing M characters of either 0 or 1 representing the binary grid you constructed. If multiple construction exists, you may output any of them. If such a grid does not exist, output -1.

Sample Input 1	Sample Output 1
6 7 4 3 2 2 3 4	1010101 0101010 1010000 0000101 0101010 1010101

Sample Input 2	Sample Output 2
2 2 2 2	-1

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Problem D
Count DFS Graph

You are currently researching a graph traversal algorithm called the Depth First Search (DFS). Starting with an empty list `A`, the following pseudocode will fill the list `A` with the visitation order of a DFS algorithm.

```
DFS(v):
    append v to A
    for each u neighbour of v in ascending node number:
        if u is not in A:
            DFS(u)
```

After running `DFS(1)` from the pseudocode above, you now have a list A containing a permutation of integers from 1 to N. You now wonder how many different simple undirected graphs with N nodes there are that yield the list A that you have. Count the number, modulo $998\,244\,353$.

A graph is simple when there are no self-loops and there is at most one edge connecting each pair of nodes. Two graphs are considered different if there is an edge connecting a pair of nodes in one graph but not the other.

Input

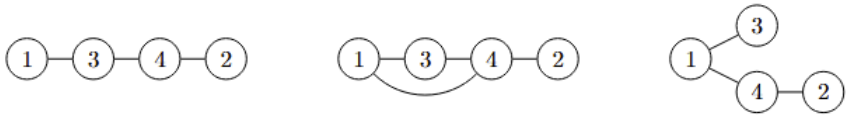
The first line contains an integer N ($2 \leq N \leq 300$). The second line contains a permutation of the first N positive integers, representing the list A. The first element of A is guaranteed to be 1.

Output

A single integer representing the number of different graphs, whose DFS order gives you the list A, modulo $998\,244\,353$.

Sample Input 1	Sample Output 1
4 1 3 4 2	3

Explanation of Sample 1: The following illustrates all the graphs with the given DFS order.



Sample Input 2	Sample Output 2
10 1 2 3 4 5 6 7 8 9 10	515546413

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Problem E
Mugen E

You are given K strings $S_1, S_2, \ldots, S_K$ of lowercase Latin letters. You are also given a string T.

Define a function $f(n)$, where n is an integer, as the following.

1. Initially, you have an empty string U.
2. For n times, choose a string uniformly at random among $S_1, S_2, \ldots, S_K$, and append that string to U.
3. Let E_n be the expected value of the number of occurrences of T in U as a substring. Then, $f(n) = E_n/n$.

It can be proven that $\lim_{n\to\infty} f(n)$ exists and can be written as a rational number p/q. Find $pq^{-1} \bmod 998\,244\,353$.

Input

The first line contains the string T ($1 \le |T| \le 5000$) consisting of lowercase Latin letters. The second line contains an integer K ($1 \le K \le 5000$). Each of the next K lines contains S_i ($1 \le |S_i| \le 5000$) consisting of lowercase Latin letters. The sum of $|S_i|$ does not exceed $1\,000\,000$.

Output

Output the limit $pq^{-1} \bmod 998\,244\,353$.

Sample Input 1	Sample Output 1
ab 2 a b	748683265

Explanation of Sample 1: It can be shown that $\lim_{n\to\infty} f(n) = \frac{1}{4}$.

Sample Input 2	Sample Output 2
ab 4 aaa abab baba bbb	1

Explanation of Sample 2: It can be shown that $\lim_{n\to\infty} f(n) = 1$.

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Problem F
Two Sets

You are given an undirected graph with N vertices and M edges, with vertices numbered from 1 to N.

You have to select two integers p and q such that:

- $N < (p+1)(q+1)$
- There exists a non-empty set of vertices S_1 such that for every vertex $u \in S_1$, there are **at least** p other vertices in S_1 that share an edge with u.
- There exists a set of vertices S_2 of size **at least** q such that for every vertex $u \in S_2$, there are no vertices in S_2 that share an edge with u.

You have to find p, q, along with any S_1 and S_2 that satisfy the requirements above. It can be proven that such p and q always exist.

Input

The first line contains two integers N and M ($1 \le N \le 2000$; $0 \le M \le \min(\frac{N(N-1)}{2}, 200\,000)$). Each of the next M lines contains two integers u and v ($1 \le u < v \le N$) representing the two vertex numbers that are connected by an edge. All the given edges are different.

Output

The first line contains two integers p and q. The second line contains an integer $|S_1|$ followed by $|S_1|$ integers representing the vertex numbers in S_1. The third line contains an integer $|S_2|$ followed by $|S_2|$ integers representing the vertex numbers in S_2.

Sample Input 1	Sample Output 1
4 2 1 2 3 4	1 2 2 1 2 2 3 1

Explanation of Sample 1: You selected $p = 1$, $q = 2$, $S_1 = \{1, 2\}$, and $S_2 = \{1, 3\}$.

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Problem G
Grid Game 2×2

There is a $10^9 \times 10^9$ grid, with rows numbered from 1 to 10^9 and columns numbered from 1 to 10^9. Denote (r, c) as the cell at row r and column c.

Initially, exactly N of the cells are black, and the rest are white. The i-th black cell is located at (R_i, C_i).

Your childhood friends, Kita and Kami, will alternately take turns playing on this grid, with Kita moving first. A turn in the game proceeds as follows.

1. Pick a black cell (r, c).
2. Pick a subset $K \subseteq \{1, 2, \ldots, t\}$ where t is the largest integer such that $2^t \leq \min(r, c)$. The set K may be empty.
3. For each $k \in K$, toggle the colour of **all** cells $(r - i, c - j)$ satisfying $0 \leq i < 2^k$, $0 \leq j < 2^k$, and $(i, j) \neq (0, 0)$. Toggling a colour means changing black colour to white and vice versa.
4. Toggle the colour of cell (r, c). Note that the colour of cell (r, c) becomes white after the toggle.

A player who is unable to play on their turn, i.e., there are no black cells anymore, loses the game, and the opposing player wins the game. If both players play optimally, determine who will win the game.

Input

The first line contains an integer N ($1 \leq N \leq 200\,000$) representing the number of black cells. Each of the next N lines contains two integers R_i and C_i ($1 \leq R_i, C_i \leq 10^9$) representing the location of the i-th black cell. All the given black cells are different.

Output

Output one line, containing either `Kita` if the first player will win the game, or `Kami` otherwise.

Sample Input 1	Sample Output 1
4 1 2 1 3 2 2 2 3	Kita

Explanation of Sample 1: The first player picks the cell $(2, 3)$ and the set $K = \{1\}$ and all the cells become white afterwards.

Sample Input 2	Sample Output 2
1 8 8	Kita

Explanation of Sample 2: The first player picks the cell $(8, 8)$ and the set K as an empty set, and the only black cell becomes white afterwards.

Sample Input 3	Sample Output 3
2 2 4 4 2	Kami

Explanation of Sample 3: The second player can always mirror the last move made by the first player.

Problem H
Maximeter

Solve the problem below for T test cases.

You are given two integers M and D. You are interested in a rooted weighted tree with the following conditions.

- Each edge has a weight of a positive integer.
- For each vertex v of the tree, there exists **no** set of v's children of size **strictly greater** than M such that all the edges connecting v and this set of children all have the same weight.
- The diameter of the tree is not greater than D. The diameter of a tree is the maximum distance between any two vertices.

Find the maximum number of vertices of such a tree. As the number of vertices can be very large, find the vertex count modulo $998\,244\,353$.

Input

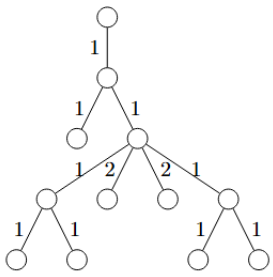
The first line contains an integer T ($1 \le T \le 100$), the number of test cases. Each of the next T lines contains two integers M and D ($1 \le M, D \le 10^9$) representing a case you have to solve.

Output

For each of the T test cases, output a single line containing the maximum number of vertices modulo $998\,244\,353$.

Sample Input 1	Sample Output 1
3 2 4 165 1 20 20	12 2 891869870

Explanation of Sample 1: The following illustrates, for the first case, a rooted tree with the maximum number of vertices.



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Problem I
International Olympiad in ICPC

Your university is hosting a competition called the IOI. As an event organiser, you want to create a banner for it. The banner you are preparing can be represented as a $3 \times N$ grid, where each cell will be printed either white or black. Due to printing issues, some of the cells cannot be printed in black.

You start with a white banner, and want to print the word `IOI` on the banner as follows.

- Form the first letter `I` by printing a **solid** rectangle of size $3 \times p$ ($p \geq 1$) in black.
- Form the letter `O` by printing a rectangular **boundary** of size $3 \times q$ ($q \geq 3$) in black.
- Form the second letter `I` by printing a **solid** rectangle of size $3 \times r$ ($r \geq 1$) in black.

The letter `O` must be formed between the letters `I`, and there must be **at least** one column between the letters. It is also required that the width of the letter `O` is **at least** the sum of the widths of the letters `I`, i.e. $q \geq p + r$. All other cells not part of the word `IOI` must remain white. Determine the maximum number of cells that you can print in black, or tell that it's impossible to print the word `IOI`.

Input

The first line contains an integer N ($1 \leq N \leq 200\ 000$), the size of the $3 \times N$ grid. Each of the next three lines contains N characters representing the cells of the grid. Each of the $3N$ characters corresponds to a cell and is either `.` or `#`, meaning you can or cannot print the cell in black, respectively.

Output

A single line representing the maximum number of cells that you can print in black. If it's impossible to print the word `IOI`, output -1 instead.

Sample Input 1	Sample Output 1
7	14

Explanation of Sample 1: You can print your banner in the following way.

```
I.OOO.I
I.O.O.I
I.OOO.I
```

Sample Input 2	Sample Output 2
6	-1

Sample Input 3	Sample Output 3
12 ...#....#... ...#.##.#... ...#....#...	22

Explanation of Sample 3: You can print your banner in the following way.

```
..I#OOOO#III
..I#O##O#III
..I#OOOO#III
```

Sample Input 4	Sample Output 4
20 #...##..... #...##.##....#...#.. #...##.....	39

Explanation of Sample 4: You can print your banner in the following way.

```
#III##OOOOOOOOOO.II
#III##O##....#..O#II
#III##OOOOOOOOOO.II
```

Problem J
AND and/or OR

Suppose you have a non-negative integer x. You can do two types of operations:

1. $x := x \text{ AND } 2x$;
2. $x := x \text{ OR } 2x$;

where AND and OR are the bitwise AND and bitwise OR operations, respectively.

You are given three integers N, A, and B.

If the value of x is initially N, is there any sequence of operations that consists of **exactly** A operations of type 1 and **exactly** B operations of type 2, such that the final value of x is $N \cdot 2^k$ for some non-negative integer k?

Input

Input consists of three integers N, A, and B ($1 \le N \le 10^{18}$, $0 \le A, B \le 10^{18}$, $A + B \ge 1$).

Output

Output a single line containing `YES` if it is possible to make the final value of x equal to $N \cdot 2^k$ where k is a non-negative integer, or `NO` otherwise.

Sample Input 1	Sample Output 1
14 2 2	YES

Explanation of Sample 1: Initially, $x = 14$. You can do the following sequence of operations:

1. Do a type 1 operation. $x = 14 \text{ AND } 28 = 12$.
2. Do a type 1 operation. $x = 12 \text{ AND } 24 = 8$.
3. Do a type 2 operation. $x = 8 \text{ OR } 16 = 24$.
4. Do a type 2 operation. $x = 24 \text{ OR } 48 = 56$.

The final value of x is $56 = 14 \cdot 2^2$.

Sample Input 2	Sample Output 2
1 0 9	NO

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Problem K
Travelling Taro Trains

Taro is travelling within a country, which consists of N cities numbered from 1 to N. There are also K train companies, numbered from 1 to K, that operate the trains within the nation. Initially, there are M one-way train services, each of which can be described as a triple (u, v, k), meaning that there are trains operated by company k that goes from city u to city v.

Taro is currently in city 1. In the next Q months, one of the following three events may happen.

- `1 u v k`: company k **starts** a new train service that goes from city u to city v. In other words, add (u, v, k) to the train network.
- `2 u v k`: company k **ends** its train service that goes from city u to city v. In other words, remove (u, v, k) from the train network.
- `3 k`: Taro either **stays** in the city he is currently in, or **takes** one of the trains operated by company k from the city he is currently in to another city.

It is guaranteed to you that the existing services (u, v, k) are always distinct throughout the course of events, and event 2 always removes a currently existing service.

Every time an event 3 happens, find the number of different cities Taro can possibly be in, given the course of events so far.

Input

The first line contains three integers N, M, K, and Q ($2 \le N \le 300\,000$; $1 \le M, K, Q \le 300\,000$). Each of the next M lines contains three integers u, v, k ($1 \le u, v \le N$; $1 \le k \le K$; $u \ne v$) describing the initial train services.

Each of the next Q lines contains an event as described above. The input guarantees that the existing services (u, v, k) are always distinct throughout the event.

Output

For each event 3, output, in a single line, the number of different cities Taro can possibly be in.

Sample Input 1	Sample Output 1
6 4 2 5 1 3 1 3 6 2 3 2 2 3 5 2 3 2 3 1 1 1 4 2 2 3 6 2 3 2	1 2 5

Explanation of Sample 1: Initially, Taro is in city 1.

1. In the first event 3, there are no train services operated by company 2 that departs from city 1, so Taro has to stay in city 1.
2. In the second event 3, Taro can either stay in city 1, or take the service $(1, 3, 1)$ and end in city 3. Because Taro can be in either city 1 or 3, you have to output 2.
3. In the third event 3, the existing train services are $(1, 3, 1)$, $(1, 4, 2)$, $(3, 2, 2)$, $(3, 5, 2)$. If Taro is currently in city 1, he can take service $(1, 4, 2)$ and end in city 4. If Taro is currently in city 3, he can take services $(3, 2, 2)$ or $(3, 5, 2)$ and end in city 2 or 5 respectively. Since he can possibly in city 1, 2, 3, 4, and 5, you have to output 5.

Problem L
Down Right Chips

You have a board of size $N \times M$, with rows numbered from 1 to N from top to bottom, and columns numbered from 1 to M from left to right. Initially, there are $A_{i,j}$ chips on the tile at the i-th row and j-th column.

You will play a solitaire game on this board. In one move, you can do one of the following.

- Choose a tile not in the bottommost row with at least two chips on it. Discard one chip from the tile, and move another chip to the tile directly below it.
- Choose a tile not in the rightmost column with at least one chip on it. Move one chip from the tile to the tile directly to its right.

The game ends when there are no more moves you can make.

There will be Q changes to the board, and each of the changes will **add** W chips to the tile at the X-th row and Y-th column. After each change, calculate the minimum number of moves to end the game using the current board.

Input

The first line contains three integers N, M, and Q ($1 \le N \le 5$; $1 \le M, Q \le 100\,000$). The next N lines, each containing M integers, represent $A_{i,j}$ ($0 \le A_{i,j} \le 100\,000$). Each of the next Q lines contains three integers X, Y, and W ($1 \le X \le N$; $1 \le Y \le M$; $1 \le W \le 10\,000$) representing a change to the board.

Output

Output Q lines, each containing a single integer representing the minimum number of moves to end the game after each change.

Sample Input 1	Sample Output 1
2 3 2 0 2 1 0 1 1 1 2 1 2 3 5	5 5

Explanation of Sample 1: After the first change, you can play as follows for the minimum number of moves.

1. Move *right* on tile $(1, 2)$.
2. Move *down* on tile $(1, 2)$.
3. Move *right* on tile $(2, 2)$.
4. Move *right* on tile $(2, 2)$.
5. Move *down* on tile $(1, 3)$.

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Problem M
Predisposed

You are given two integers N and M. You are also given Q constraints in the form (X_j, Y_j).

A sequence $A = (A_1, \ldots, A_N)$ of length N is *predisposed* if and only if:

- For each $1 \leq i \leq N$, it holds that $0 \leq A_i < M$.
- For each $1 \leq j \leq Q$, the sum of all A_k where k is a multiple of X_j is equal to Y_j under modulo M. Formally, $\sum_{X_j | k} A_k \equiv Y_j \pmod{M}$.

Find the number of all possible predisposed sequences. As the answer can be quite large, compute it modulo $998\,244\,353$.

Input

The first line contains three integers N, M, and Q ($1 \leq N, M < 998\,244\,353$; $1 \leq Q \leq \min(N, 100\,000)$). Each of the next Q lines contains two integers X_j and Y_j ($1 \leq X_j \leq N$; $0 \leq Y_j < M$; $X_p \neq X_q$ for $p \neq q$) representing a constraint.

Output

Output an integer in a single line representing the number of predisposed sequences modulo $998\,244\,353$.

Sample Input 1	Sample Output 1
2 3 2 1 0 2 1	1

Explanation of Sample 1: The only predisposed sequence is $A = (2, 1)$.

Sample Input 2	Sample Output 2
100000 100000 1 100000 0	373304036

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